Learning Earth Observation Embeddings with Temporal and Non-Temporal Self-Supervision using Random Convolutional Features

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Abstract. Self-supervised learning (SSL) has become a critical paradigm in Earth Observation (EO) due to the high cost of manual labelling. This work uses a lightweight alternative that combines Random Convolutional Features (RCF) with an SSL objective to learn temporally aligned embeddings in a fixed random feature space. We treat the RCF encoder as a frozen backbone to maintain efficiency and focus on the efficacy of representation learning strategies. We introduce and compare two primary methods: a non-temporal baseline that treats seasonal observations as unordered views for invariance learning, and a temporal variant that explicitly models seasonal consistency through a learned temporal attention module. Both models were trained on the complete SSL4EO-S12 v1.1 data set, and evaluations were conducted on the NeuCoBench benchmark, which covers a diverse set of regression and classification tasks, including biomass estimation, land cover mapping, and cloud detection. The temporal variant achieved an overall score of 18.54, which was competitive with the non-temporal variant's 18.09. Notably, our lightweight approach outperforms recent work on NeuCoBench tasks.

Keywords: Self-Supervised Learning, Random Convolutional Features, Neural Embedding

1 Introduction

The rapid expansion of Earth Observation (EO) constellations, such as Sentinel-1 and Sentinel-2, provides a continuous stream of multi-modal and multi-temporal data essential for monitoring global environmental dynamics. However, the utility of this vast data repository is frequently constrained by the high cost and logistical difficulty of obtaining high-quality ground-truth labels. Consequently,

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self-supervised learning (SSL) has emerged as a fundamental paradigm for learning task-agnostic representations from unlabeled satellite imagery. By leveraging the inherent structure of EO data, SSL models can capture complex geophysical features that transfer effectively to downstream applications.

Current advancements in EO-SSL are primarily dominated by high-parameter architectures, specifically Vision Transformers (ViTs) pretrained through masked autoencoding or contrastive objectives [1, 2]. While effective at capturing high-level semantics, these models are computationally intensive and often introduce significant storage and bandwidth bottlenecks. Furthermore, many existing frameworks treat temporal observations as independent views, thereby neglecting the seasonal consistency and phenological patterns inherent in multi-temporal satellite time series. This lack of explicit temporal modelling can limit a model's ability to distinguish between transient noise and meaningful seasonal transitions [3].

To address these limitations, we used a lightweight alternative to EO representation learning based on Random Convolutional Features (RCF) [4], as in [5]. RCF is a training-free feature extractor that uses fixed, randomly initialised Gaussian weights to map multi-sensor inputs into a high-dimensional feature space. Unlike fully learned backbones, the RCF encoder remains frozen, significantly reducing the computational overhead and ensuring that performance gains are derived from the efficacy of the representation strategy rather than sheer model capacity. Previous studies have demonstrated that such random features are particularly adept at capturing low-level texture and spectral statistics, which are often more critical for EO regression tasks than the high-level semantic abstractions targeted by transformer-based models [6, 7].

In this work, we enhance the RCF framework [4, 5] by integrating it with a VICReg-based SSL objective [8], focusing on two distinct strategies for multi-seasonal data. We emphasise that our approach does not learn the backbone representation itself. Instead, we investigate how lightweight self-supervised objectives can align and re-weight fixed random convolutional features to improve temporal robustness. This work addresses efficiency and temporal consistency challenges in large-scale EO representation learning. Our main contributions are:

1. We show that competitive EO representations can be obtained without training a deep backbone by using a frozen Random Convolutional Feature (RCF) encoder combined with lightweight self-supervised projection heads.
2. Temporal SSL variant of our model explicitly enforces seasonal consistency through learned temporal attention and SSL objectives, improving robustness under seasonal appearance shifts.
3. We demonstrate that our approach achieves strong NeuCoBench performance, remaining competitive with other works under the standardised linear probing evaluation protocol.

2 Literature Review

Self-supervised learning (SSL) has emerged as a key paradigm for learning general-purpose representations from the massive volumes of unlabeled satellite imagery provided by missions such as Sentinel-1 and Sentinel-2 [9]. Because labelled EO data are costly, sparse, and often unevenly distributed across regions and downstream tasks, SSL has become central to scalable Earth observation modelling. Contemporary EO-SSL approaches are increasingly dominated by high-capacity Vision Transformers (ViTs), trained with objectives such as masked image modelling (ViT-MAE) or self-distillation (ViT-DINO) [1, 2]. While these foundation-scale models achieve strong performance on high-level semantic tasks (e.g., scene classification), their substantial computational demands, storage requirements, and bandwidth overhead make terabyte-scale training and repeated experimentation prohibitive in many practical settings. Moreover, they can underperform on geophysical regression problems, where low-level texture cues, spectral statistics, and fine-grained spatial details are essential [10]. These challenges motivate the development of efficient alternatives that emphasise lightweight representation extraction and embedding-centric evaluation, especially when downstream assessment spans many heterogeneous tasks rather than a single benchmark. In this context, benchmarking frameworks such as NeuCoBench [11] enable standardised multi-task comparison through compressed embeddings and linear-probing protocols [12], supported by broader progress in neural data compression for scalable representation learning [1]. Random Convolutional Features (RCF) provide an efficient, training-free alternative to fully learned convolutional networks and transformer encoders [4]. Rooted in random feature approximations of kernel methods, RCF applies fixed, randomly initialised convolutional filters to project multi-sensor EO inputs into a high-dimensional representation space. Surprisingly, this simple frozen mapping has been shown to serve as a strong baseline in hyperspectral classification and SAR target recognition, while avoiding the computational burden of end-to-end backbone training [6, 7]. Recent large-scale evaluations further suggest that frozen or weakly trained encoders can remain highly competitive on regression-heavy EO benchmarks, where preserving fine-grained spectral and textural information is critical [13]. This perspective aligns with embedding-centric and compression-aware EO pipelines, which emphasise scalable representation extraction over repeated backbone tuning [1]. Notably, the MOSAIKS framework [14, 5] demonstrates that fixed random convolutional filters combined with simple linear probing can provide surprisingly strong and scalable baselines for global satellite imagery, closely related to our RCF (No Training) setting [14].

Earth observation imagery is inherently multi-temporal, as satellite acquisitions capture seasonal and phenological dynamics over time. However, many generic SSL frameworks treat observations from different acquisition dates as independent, unordered views, thereby underutilising the structured supervisory signal provided by temporal evolution cite bib24. Recent EO-specific efforts, such as EarthPT [15], SatCLIP [16], and spatio-temporal or change-aware contrastive approaches [17], have begun to explicitly incorporate temporal in-

formation into representation learning cite bib26. Earth observation data are intrinsically multi-modal, as complementary sensors such as Sentinel-1 SAR and Sentinel-2 optical imagery capture different physical properties of the Earth's surface. Optical observations provide rich spectral and semantic information, while SAR offers weather-independent structural measurements but introduces modality-specific noise characteristics such as speckle and geometric distortions. Learning representations that remain robust across these sensing conditions is therefore a central challenge in EO representation learning[18]. Recent self-supervised approaches have explored modality-invariant objectives to align embeddings across multispectral and radar domains, aiming to reduce sensitivity to nuisance factors while preserving geophysically meaningful signals [19]. Such invariance is particularly important in multi-temporal EO settings, where variations across acquisition conditions, atmospheric effects, and seasonal dynamics can otherwise dominate learned features. Methods such as VICReg enforce embedding quality through three complementary principles: invariance across augmented views, variance preservation to avoid collapse, and decorrelation to reduce redundancy between feature dimensions [8]. Unlike contrastive approaches, these objectives avoid reliance on large negative batches and have shown strong stability in large-scale training. In Earth observation, SSL with redundancy reduction is particularly appealing due to the high spectral dimensionality, multi-modal variability, and limited labelled supervision typical of remote sensing datasets. Such objectives are well-suited for regression-oriented geophysical tasks, where preserving fine-grained spectral and textural information is essential. Motivated by these properties, our temporal SSL variant builds on VICReg by integrating lightweight seasonal attention and enforcing consistency across temporal partitions in EO datacubes. Conventional evaluation in Earth observation representation learning has often focused on performance over isolated downstream tasks, making it difficult to assess the general-purpose quality and transferability of learned embeddings across diverse geophysical objectives. The recent introduction of NeuCoBench [11] addresses this limitation by providing a standardised multi-task benchmark suite that includes regression and classification problems such as biomass estimation, land cover mapping, and urban heat island analysis. NeuCoBench evaluates representations through a task-agnostic linear probing protocol, enabling consistent comparison of embedding quality across heterogeneous EO tasks. In parallel, recent work has further emphasised the importance of embedding-centric evaluation and compression frameworks for scalable multi-task EO modelling [20].

3 Methods

The proposed methodology for this work explores a lightweight approach to Earth Observation (EO) representation learning by combining fixed-feature extractors with modern self-supervised objectives. We investigate the effectiveness of Random Convolutional Features (RCF) as a frozen backbone and introduce a temporal self-supervised learning (SSL) framework designed to capture seasonal

Table 1: Summary of notation used throughout the paper.

Symbol	Description
B	Batch size
T	Number of seasonal observations per location ($T = 4$)
C	Number of input channels ($C = 30$)
H, W	Spatial height and width of input images
$\mathbf{x}_t$	Input tensor at season t
$f(\cdot)$	Frozen Random Convolutional Feature (RCF) encoder
$\mathbf{h}_t = f(\mathbf{x}_t)$	RCF feature vector at season t
D_f	Dimension of RCF features (4096)
$g(\cdot)$	Projector network
$\mathbf{z}_t = g(\mathbf{h}_t)$	Projected embedding at season t
D_z	Embedding dimension (1024)
$\mathcal{A}(\cdot)$	Temporal attention aggregation module
w_t	Learned attention weight for season t
$\mathbf{v}$	Temporally aggregated seasonal embedding
$\mathcal{L}_{sim}$	VICReg similarity (invariance) term
$\mathcal{L}_{var}$	VICReg variance regularization term
$\mathcal{L}_{cov}$	VICReg covariance regularization term
λ, μ, ν	Coefficients weighting the VICReg components

patterns. The details of notations used are shown in Table 1. The overall pipeline is shown in Fig. 1. Stage 1 performs upstream self-supervised pre-training on the full SSL4EO-S12 v1.1 Sentinel-2 seasonal datacube using a fixed RCF feature backbone with projection and compression modules. Stage 2 converts the extracted unlabeled embeddings into downstream labelled embedding-label pairs by merging datasets and formatting them. Stage 3 evaluates representation quality using the NeuCo-Bench framework via linear probing with regression and classification heads, producing benchmark scores and rankings. Our contribution is shown in Fig. 2, where multi-modal seasonal datacubes ($T = 4$) are encoded using a frozen Random Convolutional Feature (RCF) backbone producing 4096-D representations, which are mapped into a 1024-D embedding space through a lightweight MLP projector and two variants are tried out. (a) Non-temporal variant: a single seasonal timestamp is randomly sampled, and two augmented views are aligned using the VICReg [8] objective. (b) Temporal variant: embeddings from all four seasons are randomly partitioned into two temporal subsets and aggregated via a learned temporal attention module before applying VICReg, explicitly enforcing seasonal consistency.

To extract visual representations from multimodal Earth observation data cubes, we adopt Random Convolutional Features (RCF) as our backbone encoder, as done in [5, 14]. Rather than training a deep convolutional network end-to-end, RCF provides a lightweight alternative based on fixed random kernels. Each geographic location is represented as a multi-season tensor:

$$\mathbf{x} \in \mathbb{R}^{T \times C \times H \times W},$$

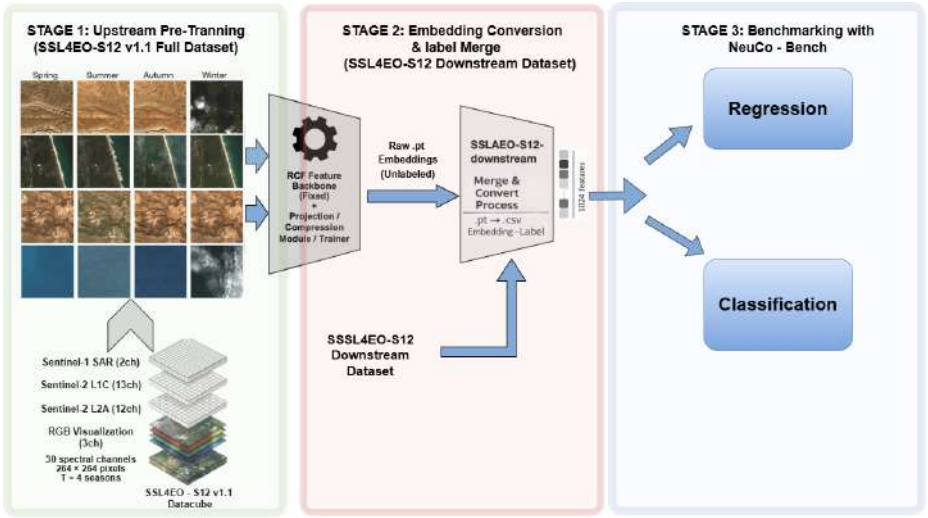


Fig. 1: Three-stage pipeline for SSL4EO-S12 embedding-based benchmarking.

where $T = 4$ corresponds to seasonal observations and $C = 30$ denotes concatenated multimodal channels. The RCF encoder applies convolutional kernels sampled once from a Gaussian distribution:

$$W \sim \mathcal{N}(0, \sigma^2),$$

and computes feature responses as:

$$\Phi(\mathbf{x}_t) = W * \mathbf{x}_t + b,$$

where b is a fixed bias term (set to -1.0 in our implementation). The resulting representation is pooled into a compact feature vector with $D_f = 4096$, i.e.,

$$f_{\text{RCF}} : \mathbb{R}^{C \times H \times W} \to \mathbb{R}^{4096}.$$

A central aspect of our method is treating RCF as a fully frozen backbone. During training, no gradients are propagated through the convolutional filters:

$$\nabla_W \mathcal{L} = 0.$$

This means that the backbone does not learn task-specific weights; instead, all improvements in representation quality arise from the subsequent projection and temporal alignment modules. Overall, this design provides an efficient way to study temporal self-supervision on top of a fixed random feature space, following the scalable philosophy introduced in MOSAIKS [14, 5]. This frozen encoder plus lightweight learning philosophy is also aligned with work showing that fixed feature extractors and embedding-centric pipelines can be strong, scalable baselines in EO settings.[20, 21]

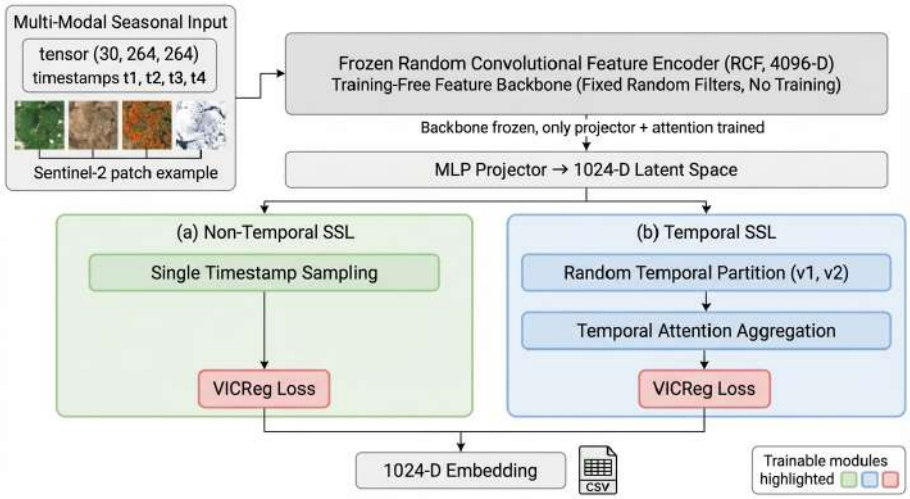


Fig. 2: Overview of the proposed RCF-VICReg self-supervised learning framework on SSL4EO-S12 v1.1.

The frozen RCF backbone produces high-dimensional feature vectors $\mathbf{h}_t \in \mathbb{R}^{4096}$ for each seasonal observation. Following standard practice in self-supervised representation learning, we introduce a learnable projection head that maps these features into a compact embedding space where the self-supervised objective is applied.

Projection Head. For each timestamp t, the RCF feature vector is projected as:

$$\mathbf{z}_t = g(\mathbf{h}_t), \qquad \mathbf{z}_t \in \mathbb{R}^{D_z},$$

where $D_z = 1024$ is the embedding dimension and $g(\cdot)$ denotes a two-layer MLP.

Specifically, the projector is defined as:

$$g = \text{Linear}(4096, 2048) \rightarrow \text{GELU} \rightarrow \text{Linear}(2048, 1024).$$

Thus, for a multiseason datacube with $T = 4$ timestamps, we obtain a sequence of projected embeddings:

$$\mathbf{Z} = \{\mathbf{z}_1, \mathbf{z}_2, \ldots, \mathbf{z}_T\}, \qquad \mathbf{Z} \in \mathbb{R}^{T \times 1024}.$$

The projector is the main trainable component in our architecture, and it adapts the fixed random features into a task-relevant embedding space.

Temporal Attention Aggregation. To incorporate seasonal structure, we further introduce a lightweight. Temporal Attention module that aggregates the per-season embeddings into a single temporally-informed representation.

Given the sequence $\{\mathbf{z}_t\}_{t=1}^{T}$, the temporal attention output is computed as:

$$\mathbf{v} = \sum_{t=1}^{T} w_t \, \mathbf{z}_t, \qquad \mathbf{v} \in \mathbb{R}^{1024},$$

where w_t denotes the learned attention weight associated with season t.

The weights are obtained using a small scoring network:

$$s_t = a(\mathbf{z}_t),$$

followed by a softmax normalization:

$$w_t = \frac{\exp(s_t)}{\sum_{k=1}^{T} \exp(s_k)}.$$

In our implementation, the attention scoring function $a(\cdot)$ is parameterized as:

$$a(\mathbf{z}_t) = \text{Linear}(1024, 128) \rightarrow \tanh \rightarrow \text{Linear}(128, 1).$$

This mechanism enables the model to assign higher importance to the most informative seasonal observations while remaining robust to temporal variability such as cloud cover or seasonal appearance shifts.

3.1 Datasets

In this work, we use two datasets: the large-scale SSL4EO-S12 v1.1 [22] dataset for self-supervised pretraining, and the SSL4EO-S12-downstream [22] dataset used for downstream evaluation in the Embed2Scale Challenge [5]. The SSL4EO-S12 v1.1 dataset contains 246,144 unique geographic locations, each observed at four seasonal timestamps. Each location is represented as a multimodal satellite datacube comprising the following modalities: Sentinel-1 GRD with VV and VH polarisations, Sentinel-2 L1C (13 bands), Sentinel-2 L2A (12 bands), and Sentinel-2 RGB visualisation (3 channels). The SSL4EO-S12-downstream dataset consists of two main components: image data and labels. The image data, which contains 13,260 multimodal seasonal data cubes of s1, s2l1c and s2l2a. Labels, provided as CSV annotation files. SSL4EO-S12-downstream provides 11 regression and classification tasks, including biomass estimation, crop fraction prediction, land-cover composition, cloud coverage assessment, nodata detection, and synthetic random baselines. These tasks enable standardised benchmarking of learned embeddings under the NeuCoBench evaluation protocol.

3.2 Experimental setup

Models are trained on the full SSL4EO-S12 dataset for 10 epochs using the AdamW optimiser (lr=1e-4, weight decay=1e-4). Batch size is 12 (adjusted for memory constraints). We use gradient clipping (norm=1.0) and skip batches with NaN losses. Checkpoints are saved per epoch. For embedding generation,

we evaluate on the downstream dataset, applying the trained projector and temporal attention to produce 1024-dimensional embeddings per cube. Embeddings are assessed on NeuCoBench [11], a benchmark with regression (biomass, clouds, heat island) and classification (crops, landcover, nodata) tasks.

4 Results

Table 2: Overall performance comparison across all evaluated embedding methods on the NeuCoBench benchmark. (Higher is better)

Method	**Epochs**	**Embedding Dim**	**Overall Score**
ViT-MAE (SSL)	10	768	9.81
ViT-DINO (SSL)	10	768	9.15
RCF (No Training)	-	4096	16.65
RCF + PCA	-	768	17.49
RCF + PCA	-	1024	17.51
RCF + PCA Whitening	-	768	12.03
RCF + PCA Whitening	-	1024	10.56
Mean Baseline (Embed2Scale)	-	1024	-0.41
Random Baseline	-	1024	-6.79
Our (Non-Temporal)	5	1024	17.16
Our (Temporal)	5	1024	17.69
Our (Non-Temporal)	10	1024	18.09
Our (Temporal)	10	1024	18.54

For regression tasks, a linear regression model is trained to predict continuous target variables, biomass, cloud fraction, and heat island intensity from the embeddings. For classification tasks, a linear classifier with a softmax activation is trained to predict binary class labels (land-cover presence) and NoData.For each task t, NeuCo-Bench aggregates the K split-wise scores $s_{t,k}$ into a single quality score Q_t that accounts for both mean performance and stability:

$$Q_t = \frac{100\,\langle s_{t,k}\rangle_k}{\mathrm{std}_k(s_{t,k}) + \epsilon},$$

where $\langle\cdot\rangle_k$ denotes averaging over splits, $\mathrm{std}_k(\cdot)$ denotes standard deviation, and ϵ is a small constant to prevent numerical instability. Higher Q_t values indicate better and more stable performance. The overall score reported in this work corresponds to the average Q_t across all downstream tasks. Table 2 represents overall score by various methods.

4.1 Regression Tasks

RCF-based methods outperform transformer-based SSL approaches on most regression-oriented NeuCoBench tasks Table 3, reflecting the importance of preserving low-level spectral and textural information. Temporal self-supervised alignment further improves robustness, with the temporal RCF + VICReg model showing clear gains in biomass and heat-island estimation.

Table 3: Performance on regression-oriented downstream tasks from the NeuCoBench benchmark. Higher scores indicate better performance.

Method	**Biomass Mean**	**Biomass Std**	**Clouds**	**Heat Mean**	**Heat Std**
Mean Baseline (Embed2Scale)	-3.34	-3.55	-4.66	-0.96	-4.34
Random Baseline	-5.42	-6.34	-9.01	-10.75	-6.29
RCF (No Training, 4096)	6.32	6.91	13.58	16.05	4.81
RCF + PCA (1024)	7.65	7.29	10.67	7.90	2.27
RCF + PCA Whitening (1024)	-0.18	-2.01	-3.70	-4.15	-4.68
ViT-MAE (SSL, 768)	6.63	5.88	4.74	13.89	2.94
ViT-DINO (SSL, 768)	9.25	9.33	3.35	17.08	5.42
Our (Non-Temporal)	7.78	6.49	11.21	14.25	4.14
Our (Temporal)	7.56	6.52	9.51	15.16	4.31

4.2 Classification Tasks

Classification performance improves with temporal modelling Table 4, as seasonal alignment reduces intra-class variability caused by phenological and illumination changes. The temporal RCF + VICReg model achieves substantial gains on NoData and land-cover tasks. Although transformer-based SSL models remain effective for learning semantic hierarchies, temporal alignment substantially narrows the gap while maintaining computational efficiency. This demonstrates that lightweight temporal self-supervision can enhance classification robustness without requiring a learned backbone.

4.3 Qualitative Analysis of Learned Embeddings

We visualise the learned representation space using PCA [12], UMAP [23], and t-SNE [24] projections, highlighting the emergence of a coherent low-dimensional manifold. These results indicate that the encoder captures nontrivial structural patterns in the embedding space, supporting its effectiveness for downstream representation-learning tasks. In fig. 3, frozen RCF has an anisotropic and scattered structure, whereas temporal SSL generates a more compact and aligned embedding space. The Non-Temporal variance exhibits some improvement but still has residual directional distortions. In fig. 4, Temporal SSL creates more continuous and organised manifolds than Frozen RCF, showing enhanced seasonal consistency (t-SNE is only demonstrated for qualitative evaluation in fig.

Table 4: Performance on classification-oriented downstream tasks from the NeuCoBench benchmark. Higher scores indicate better performance.

Method	Crops	Landcover (Agriculture)	Landcover (Forest)	NoData
Mean Baseline (Embed2Scale)	2.18	0.61	4.00	-3.55
Random Baseline	-10.61	-9.23	-10.83	-7.03
RCF (No Training, 4096)	16.82	30.96	31.06	50.86
RCF + PCA (1024)	30.55	37.97	36.68	48.52
RCF + PCA Whitening (1024)	26.33	31.36	30.79	39.98
ViT-MAE (SSL, 768)	4.07	15.61	16.67	35.45
ViT-DINO (SSL, 768)	4.36	14.90	14.15	21.55
Our (Non-Temporal)	18.25	33.05	34.76	66.46
Our (Temporal)	18.73	31.90	34.07	73.76

5). The Non-Temporal variation displays intermediate organisation. UMAP is used to conduct qualitative analysis.

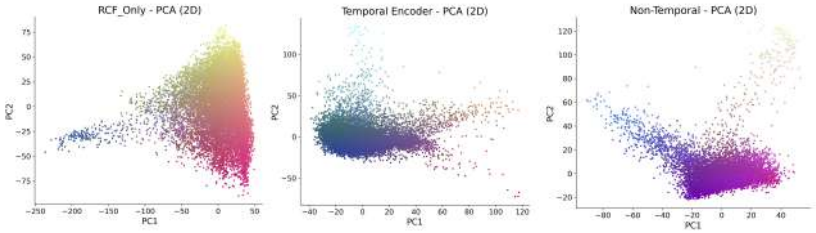


Fig. 3: PCA projection of learned embeddings (N = 13,260).(left: frozen RCF, middle: temporal SSL, right: non-temporal SSL)

5 Discussion

To better understand the contributions of various components in our framework, we conduct a series of studies. These experiments isolate the effects of temporal modelling and training duration and projector architecture, providing insights into their impact on downstream performance. An important observation from our experiments is that RCF-based representations outperform transformer-based SSL methods on regression-oriented tasks, while achieving comparatively lower gains on semantic classification objectives. This behaviour is not a limitation of the proposed framework but rather a direct consequence of the nature of Random Convolutional Features and the evaluation tasks in NeuCoBench.

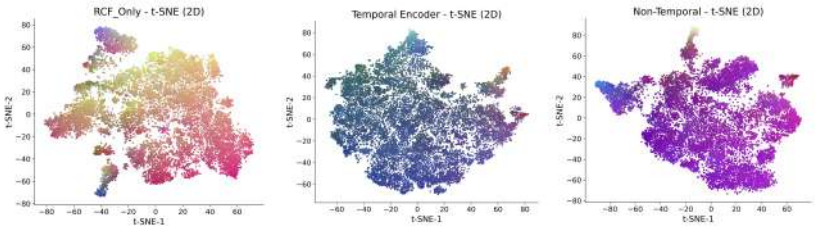


Fig. 4: t-SNE visualization of (13260) embeddings (left: frozen RCF, middle: temporal SSL, right: non-temporal SSL).

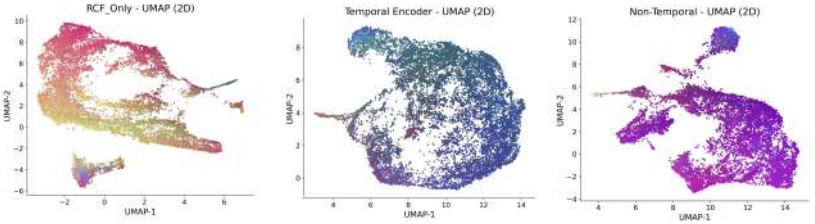


Fig. 5: UMAP visualization of (13260) embeddings (left: frozen RCF, middle: temporal SSL, right: non-temporal SSL)

Regression tasks such as biomass estimation, heat island intensity, and cloud fraction prediction depend heavily on preserving continuous physical signals, including spectral distributions, textural statistics, and radiometric consistency. The frozen RCF backbone, composed of fixed random filters, is particularly effective at capturing these low-level spectral and spatial characteristics, which are critical for geophysical regression problems. In contrast, classification tasks such as land-cover and crop-type recognition require higher-level semantic abstraction and category-specific discrimination. Transformer-based SSL models, which learn hierarchical and semantic features through large-scale optimisation, are better suited for such objectives. Since the framework intentionally avoids learning backbone parameters to improve computational efficiency, it does not explicitly optimise for semantic separability.

Nevertheless, the use of temporal self-supervision partially mitigates this gap. Explicit seasonal alignment and temporal attention improve classification performance by enforcing consistency across seasonal observations, particularly for land-cover and NoData categories. These results suggest that temporal structure provides complementary supervisory signals even when operating on fixed random features.

6 Conclusion

This study presents a self-supervised learning-based lightweight RCF-VICReg framework with temporal modelling, achieving the best NeuCoBench score. Visual results and numerical results are presented, and a detailed analysis is done. Our results suggest that under constrained compute and regression-oriented objectives, lightweight random-feature-based approaches remain competitive and merit further study alongside other models.

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